EART11100
How to
Grow a Planet



Who will lecture to you:



Prof Dunc Cameron
duncan.cameron@manchester.ac.uk
Location: John Garside Building
(Manchester Institute of Biotechnology)



Dr Rob Sansom
robert.sansom@manchester.ac.uk
Location: Michael Smith Building

When are the lectures?

Week	Date	Content	Lecturer
2	24.09	In the Beginning	Dunc
3	01.10	Oxygen	Dunc
4	08.10	Animal Origins	Rob
5	15.10	Ocean Life	Rob
6	22.10	Life moves on Land	Dunc & Rob
7	29.10	*No session – test week*	
8	5.11	Reach for the Sky	Dunc
9	12.11	Dinosaurs!	Rob
10	19.11	Wild Flowers and Fungi	Dunc
11	26.11	Cenozoic	Rob
12	03.12	Human Origins?	Rob
13?	10.12	Grasses/Humans/Future	Dunc

Information on Blackboard

- Slides will appear 24 hours before each lecture (NB 10am on Monday).
- I have also broken each lecture down into manageable sections and have written a short essay for each so you can read the content too. This will also appear at the same time as the slides and is marked as “in other words”.
- I have also added some video clips to augment the lectures and you must view these as they will help set the context for the lecture (and indeed the course).



EART11100

How to Grow a Planet

Lecture 1: In the beginning…

What is the course about?

What is Life?

Where did it come from?

How did it evolve?

Why are there lots of different species?

What do they all do?

The story of life on Earth

Learning objectives

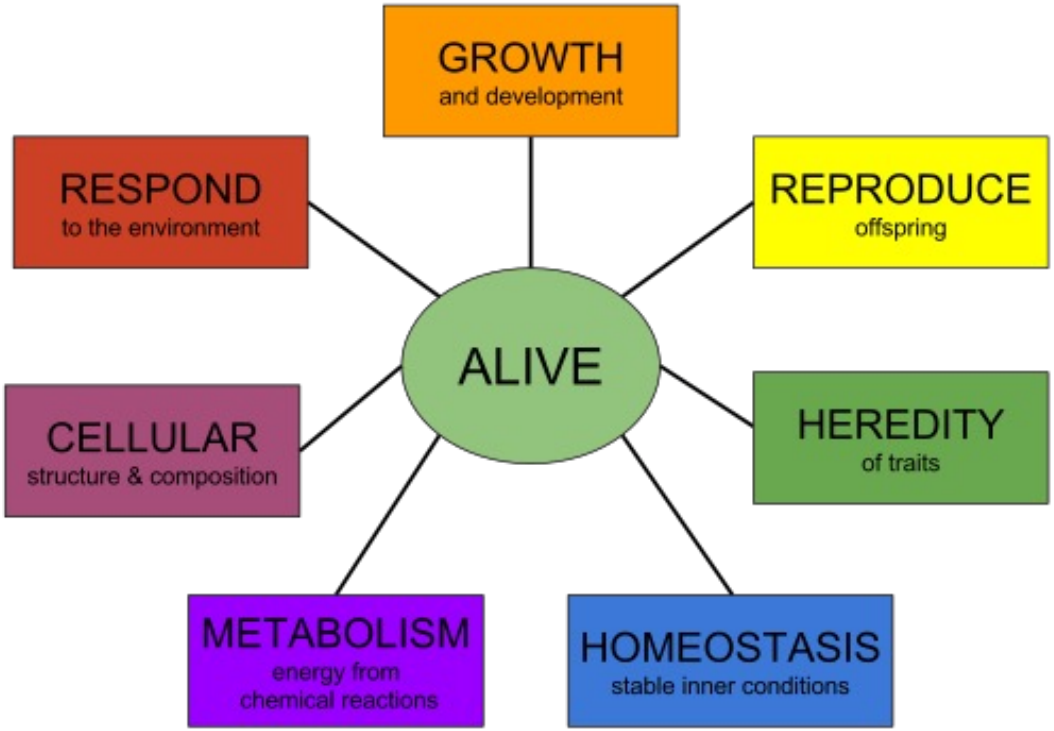
After this week's lecture and independent study you should be able to:

- Identify the main characteristics of living organisms and describe key features thought to characterise LUCA - the last universal common ancestor
- Define the major groups of organisms present on earth and explain how these are classified
- Explain that organisms use energy to make complex molecules out of simple ones
- Identify the major classes of biological molecules - proteins, carbohydrates lipids and nucleic acids
- Describe and contrast features of light capture in the archaen and bacterial lineages

LIFE!



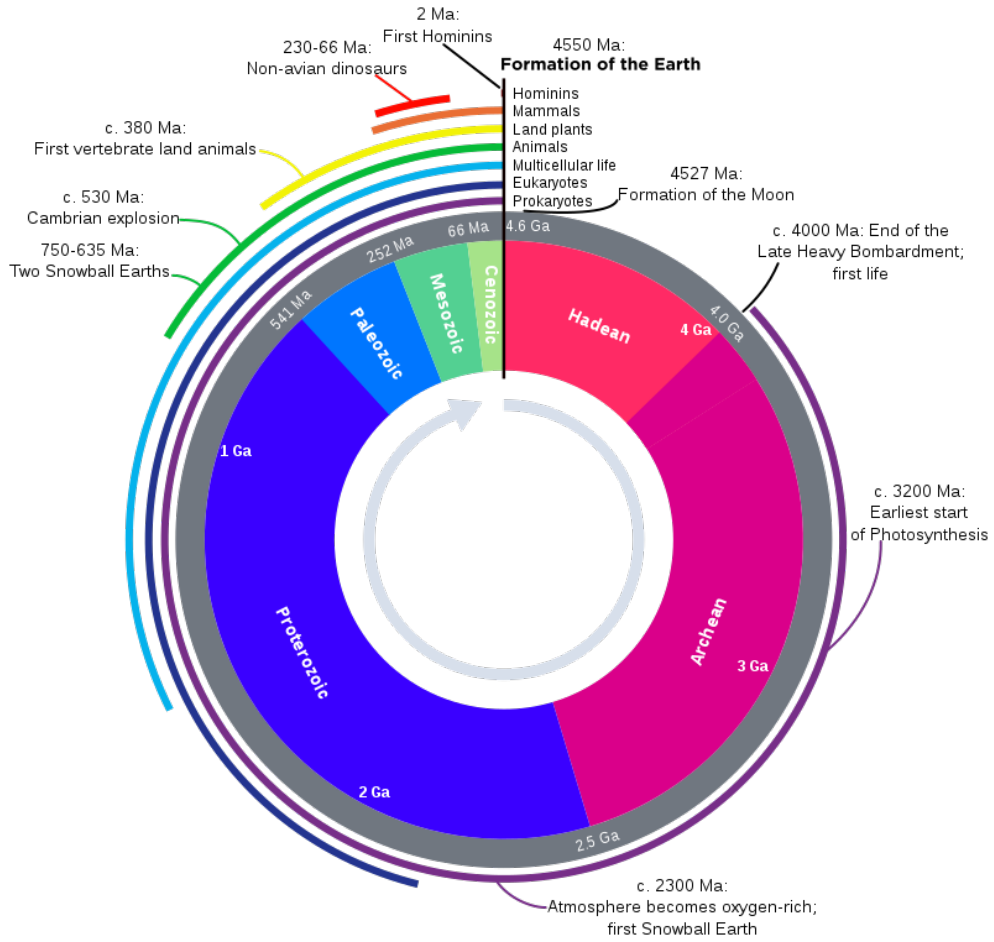
What is Life?



To be alive, a structure must possess some or all of these characters
(yes it really is that vague)

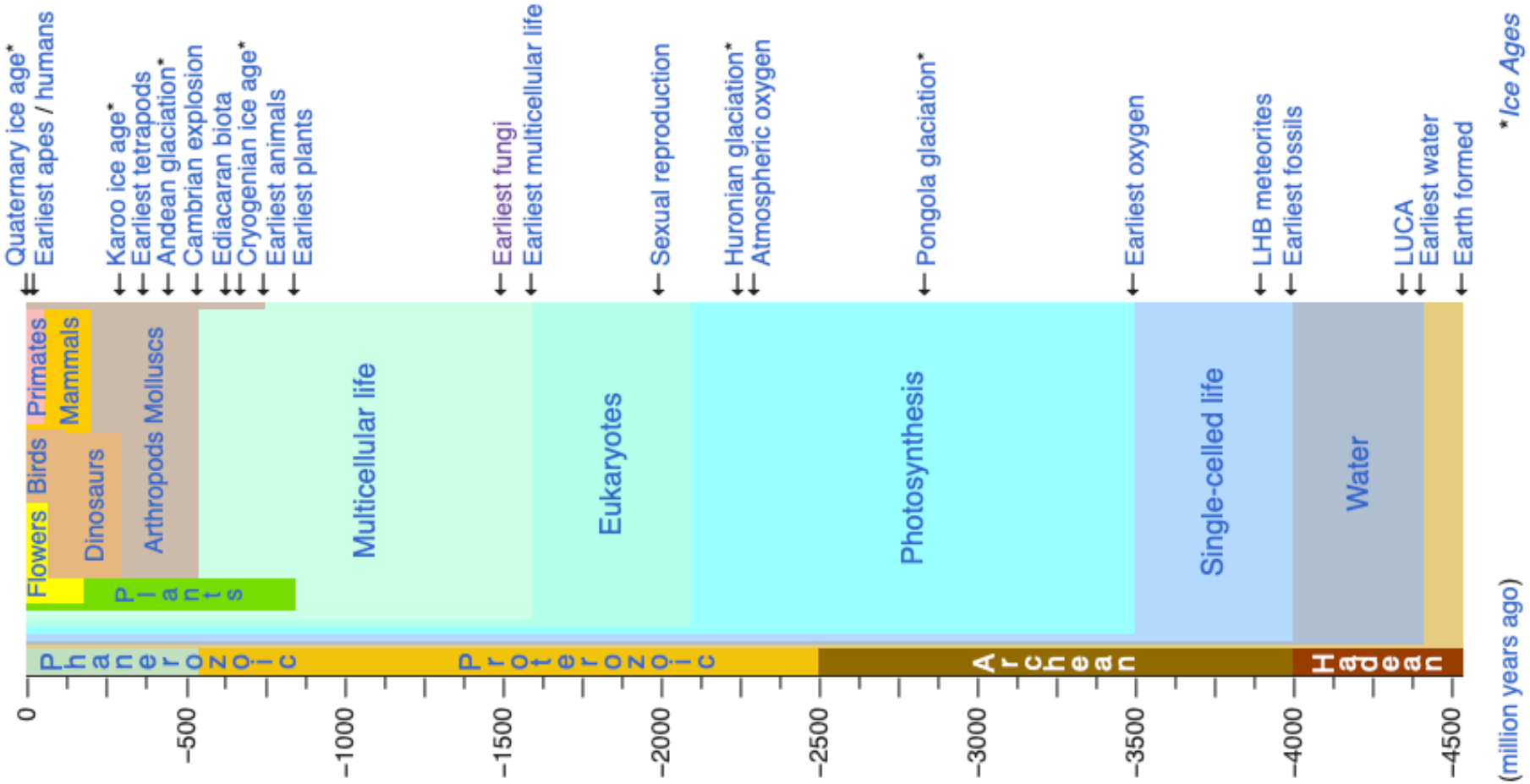
Image by Chris Packard, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=59184218
https://en.wikipedia.org/wiki/Life

The Geological Clock



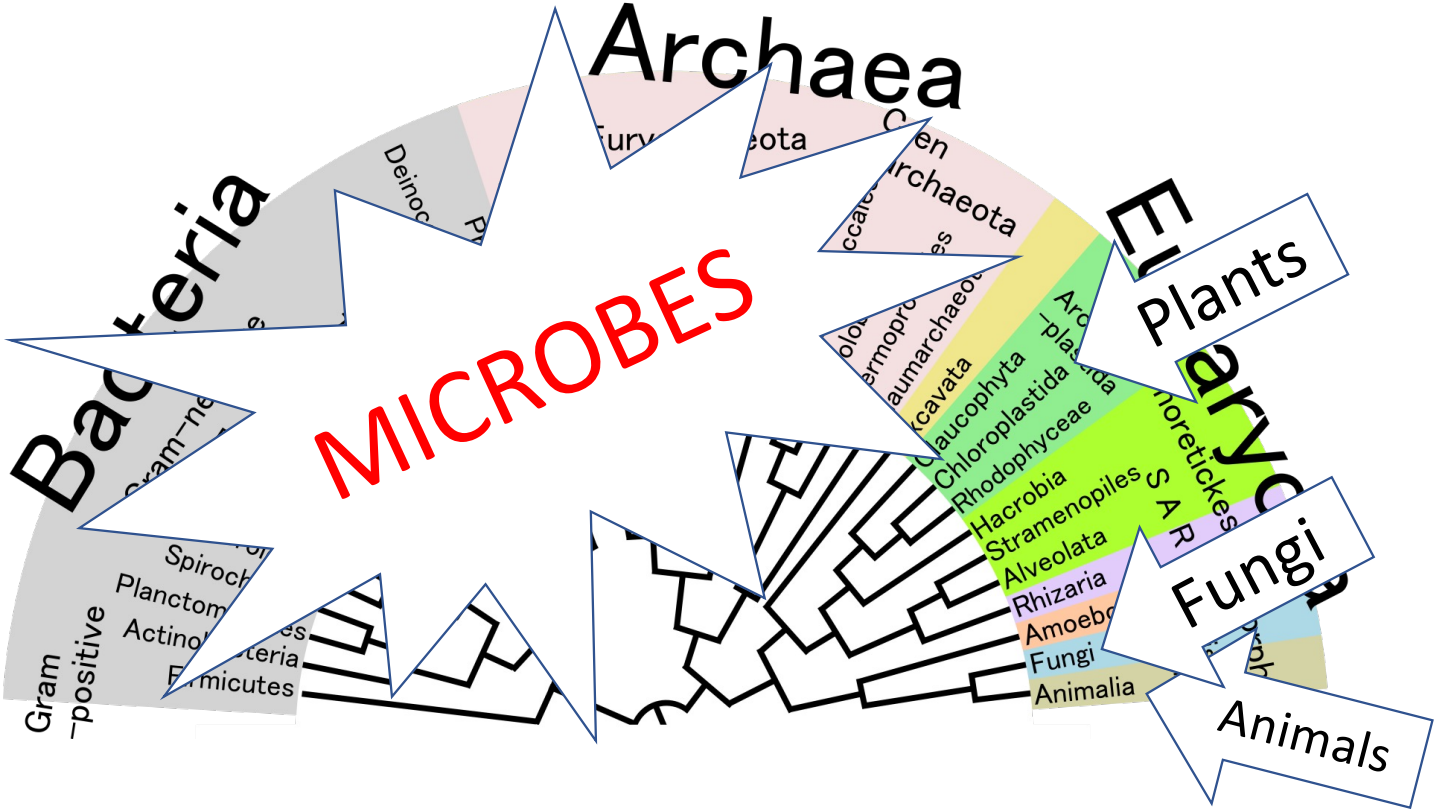
https://en.wikipedia.org/wiki/Geologic_time_scale

The Geological Clock



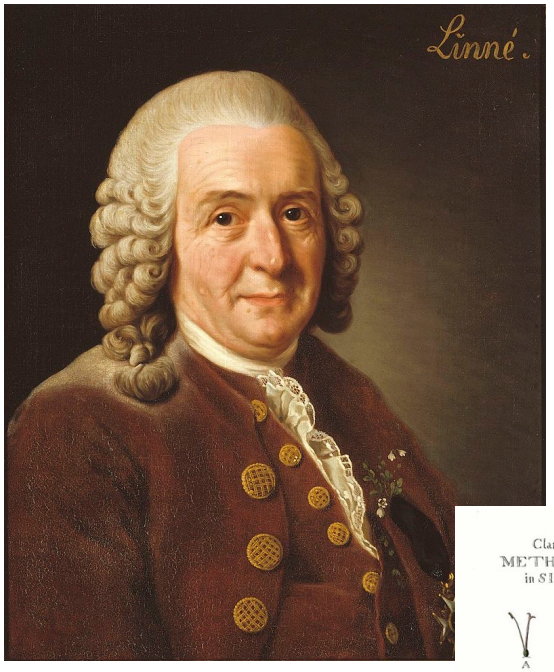
https://en.wikipedia.org/wiki/Template:Life_timeline

The tree of life…

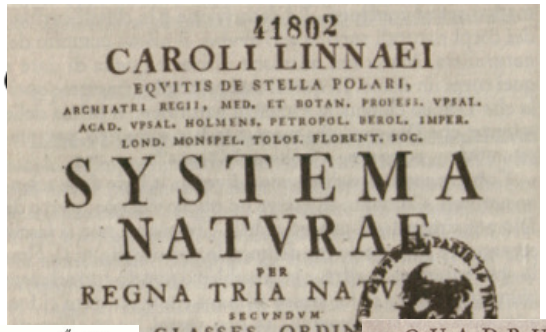


https://commons.wikimedia.org/wiki/File%3APhylogenetic_Tree_of_Life.png

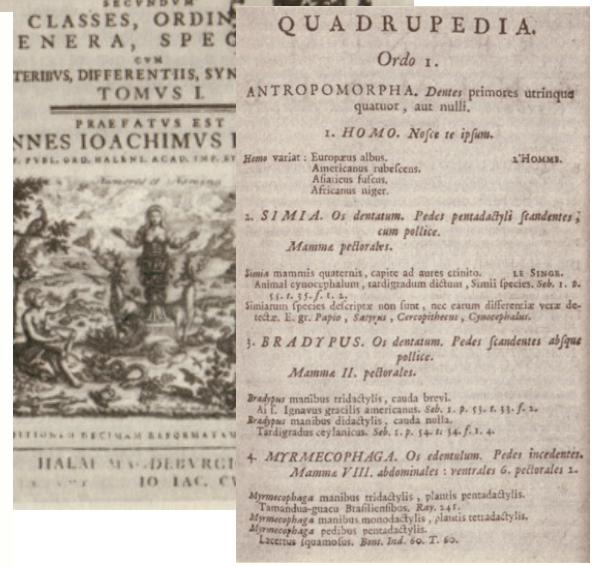
Carl Linnaeus (Carl von Linné)
1707-1778



By Alexander Roslin - Nationalmuseum Public
https://commons.wikimedia.org/w/index.php?cu







Taxonomy – the science of naming and sorting things



Digitalis purpurea L

Taxonomy – the science of naming and sorting things



Digitalis purpurea L



Digitalis minor

Taxonomy – the science of naming and sorting things



Digitalis purpurea L



Digitalis minor



Scrophularia nodosa

Taxonomy – the science of naming and sorting things



Digitalis purpurea L

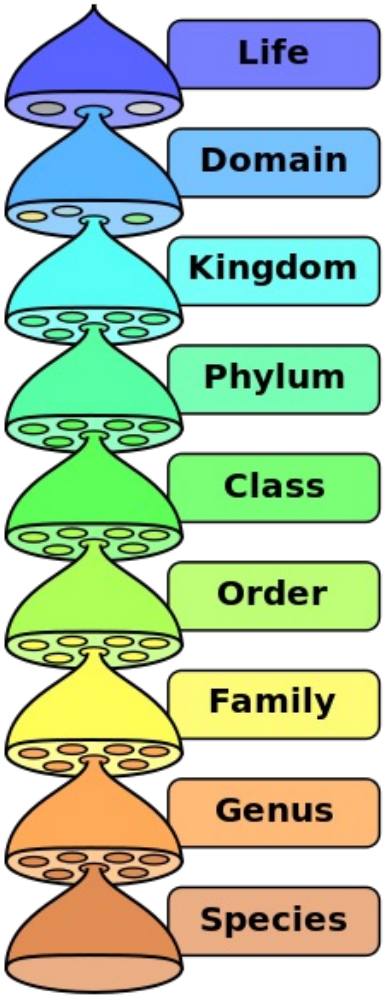


Digitalis minor



Scrophularia nodosa

Taxonomy – the science of naming and sorting things



Organisms groups by similarity

Linnaeus – Plants and Animals

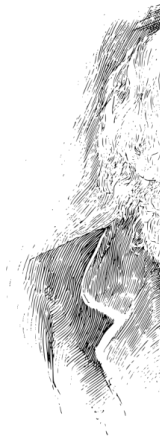
Most recent systems 3 domains and 7 kingdoms

Domains: Bacteria, Archaea, Eucarya (*n.b. NOT prokaryotes*)

Kingdoms: Bacteria-Archaea-Protista-Chromista-Plantae-Fungi-Animalia (but this could change…)

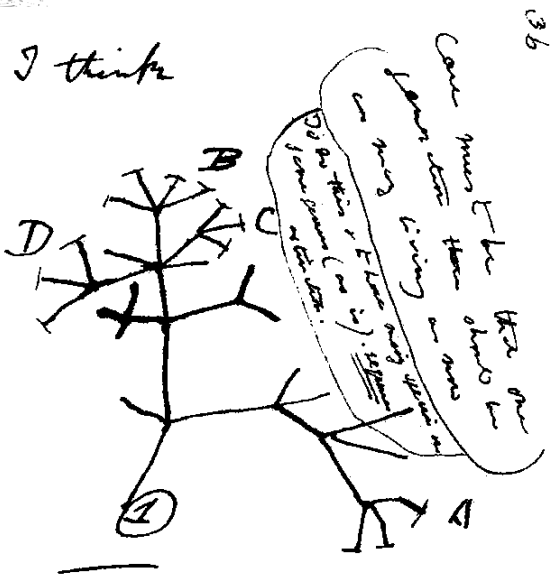
https://en.wikipedia.org/wiki/Taxonomy_(biology)
https://commons.wikimedia.org/wiki/File%3ABiological_classification_L_Pengo_vflip.svg

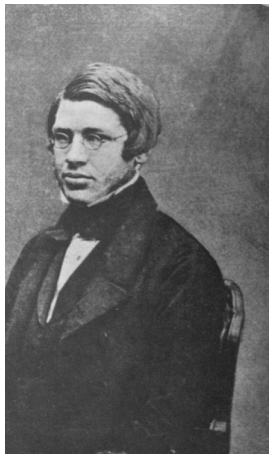
What Linnaeus failed to see…



Charl

https://commons.wikimed



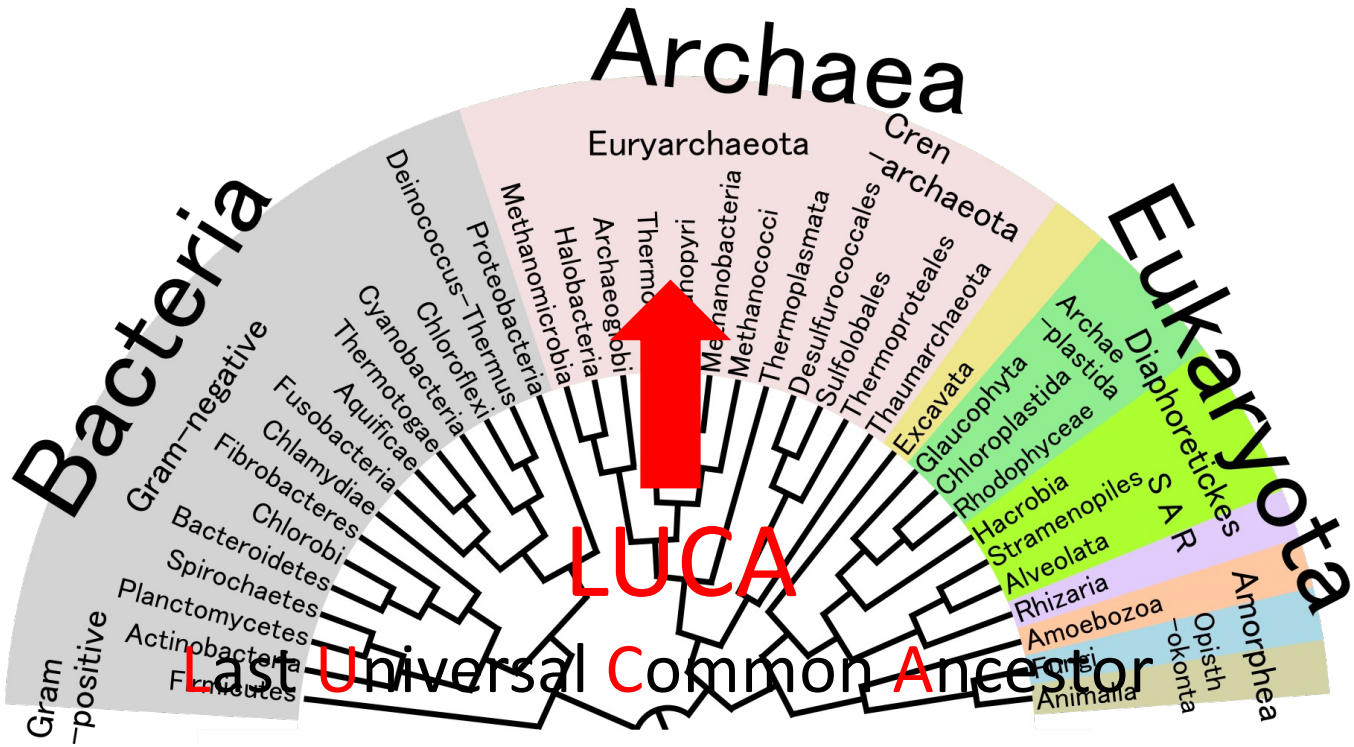


d Russel Wallace

.wikimedia.org/w/index.php?curid=841445

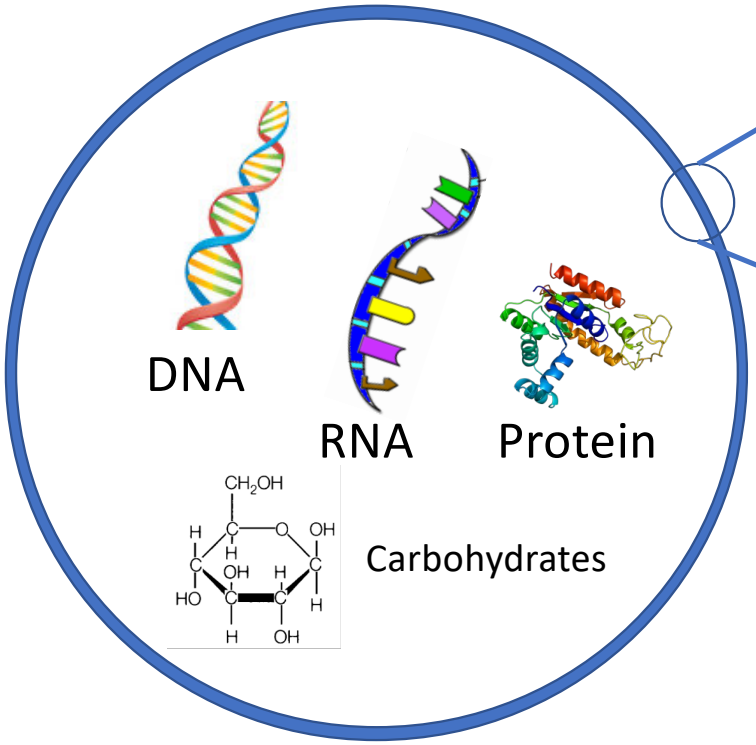
Page from Darwin's notebook, July 1873

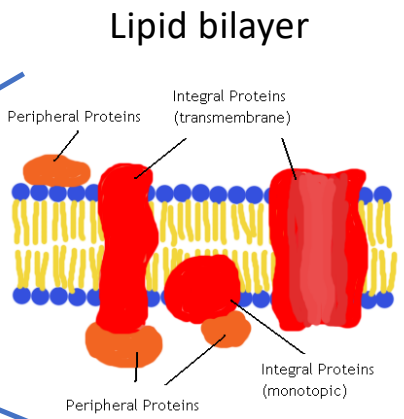
The tree of life…



All life on earth is descended from a single individual cell which possessed the characteristics shared by all living organisms today

What did LUCA look like?



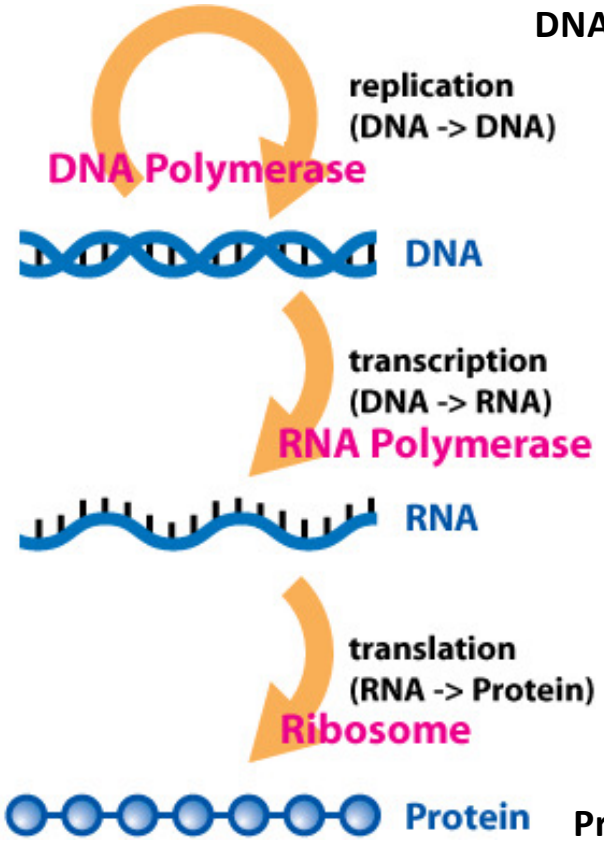


Membranes: separate the cell contents from the rest of the universe

Transport proteins: control entry and exit

Central Dogma: DNA makes RNA makes PROTEIN

The Central Dogma



DNA: chain of *nucleotides* formed into *double helix*

Codons: combinations of 3 nucleotides code for 20 amino acids

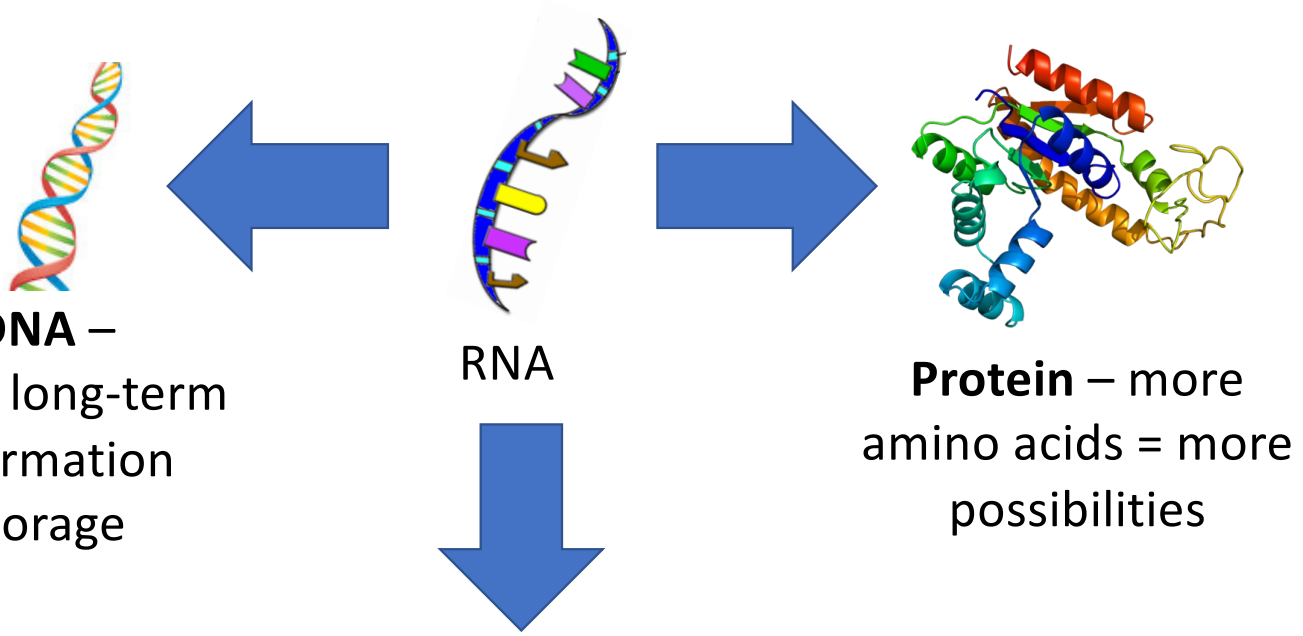
RNA: single strand; acts as a "translator" of the DNA code

Protein: chain of *amino acids*

What came first? DNA, RNA or PROTEIN?

https://en.wikipedia.org/wiki/Central_dogma_of_molecular_biology

RNA World?



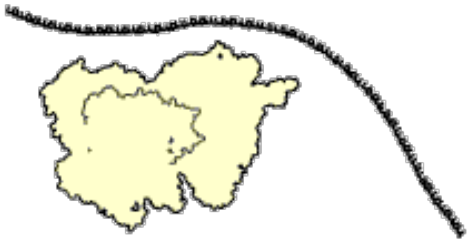
DNA – *Stable* long-term information storage

RNA

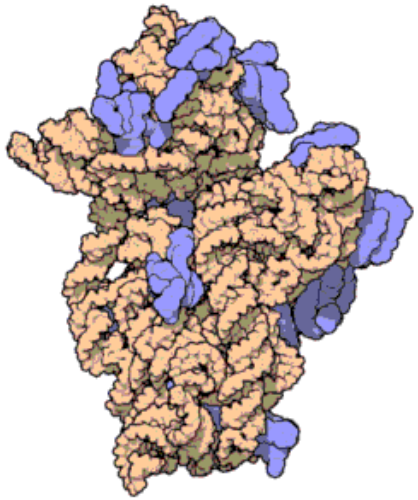
Protein – more amino acids = more possibilities

Ribozymes – catalysts made of RNA

Ribosomes

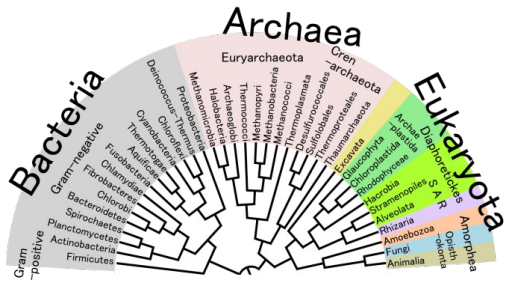


Convert genetic code into protein sequence



Made of **RNA** and **Protein**

Changes in 16S (Bacteria and Archea) and 18S (Everything else) ribosomal RNA used to construct phylogenetic trees



The second law of thermodynamics

The total entropy of an isolated system can only increase over time. (Wikipedia)

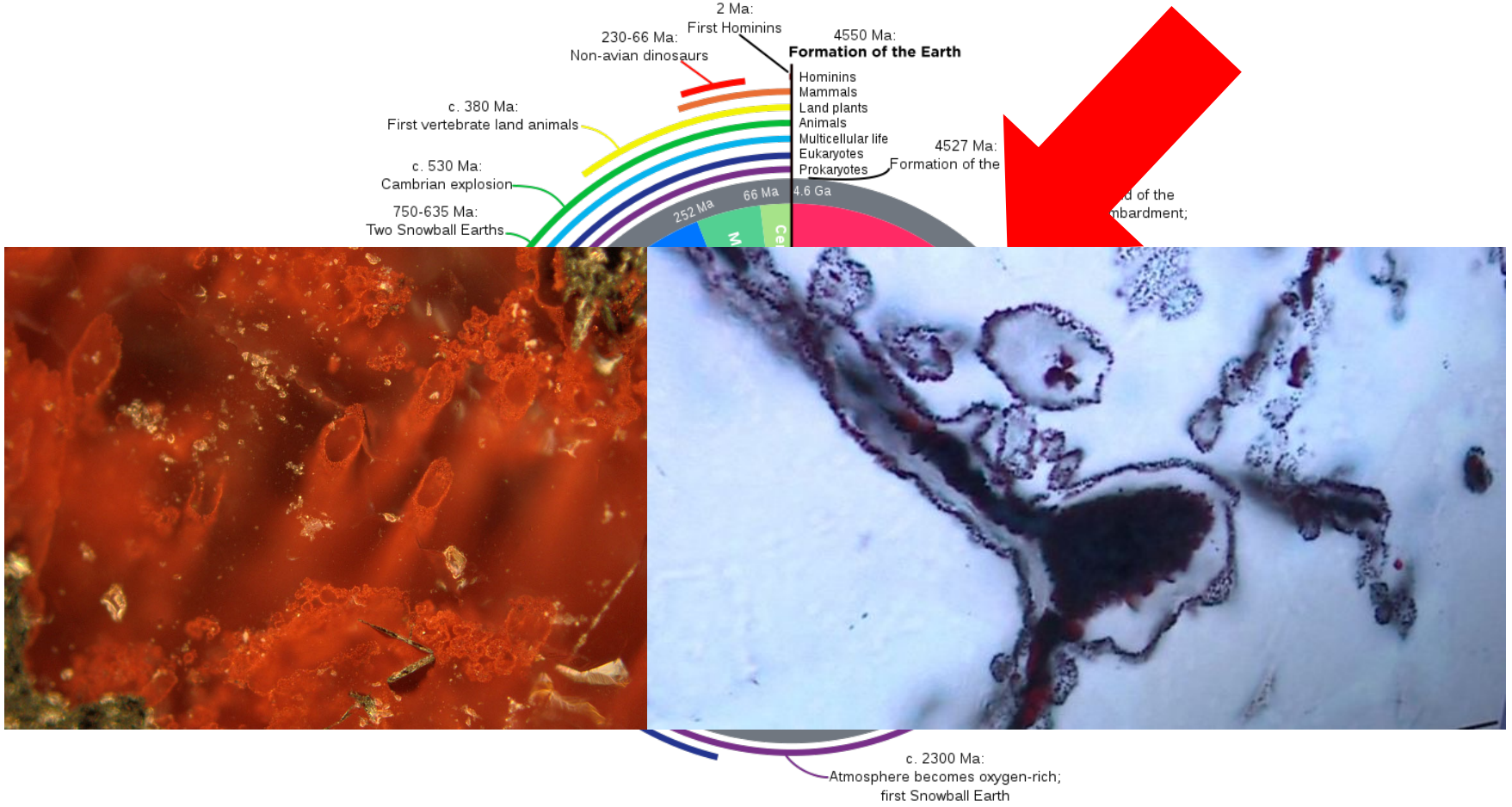
All systems tend from a state of order to a state of disorder

Things tend to fall apart

Life is a highly *ordered* process. How can life maintain and reproduce itself?

ENERGY


Pause



Dodd et al. (2017) “Evidence for early life in Earth’s oldest hydrothermal vent precipitates” Nature 543, 60–64
See http://www.ucl.ac.uk/news/news-articles/0217/010317-Worlds-oldest-fossils-unearthed

But see also: https://www.theguardian.com/science/2017/sep/27/carbon-found-in-395bn-year-old-rocks-is-remnants-of-ancient-life-say-researchers > life older than 4bn years

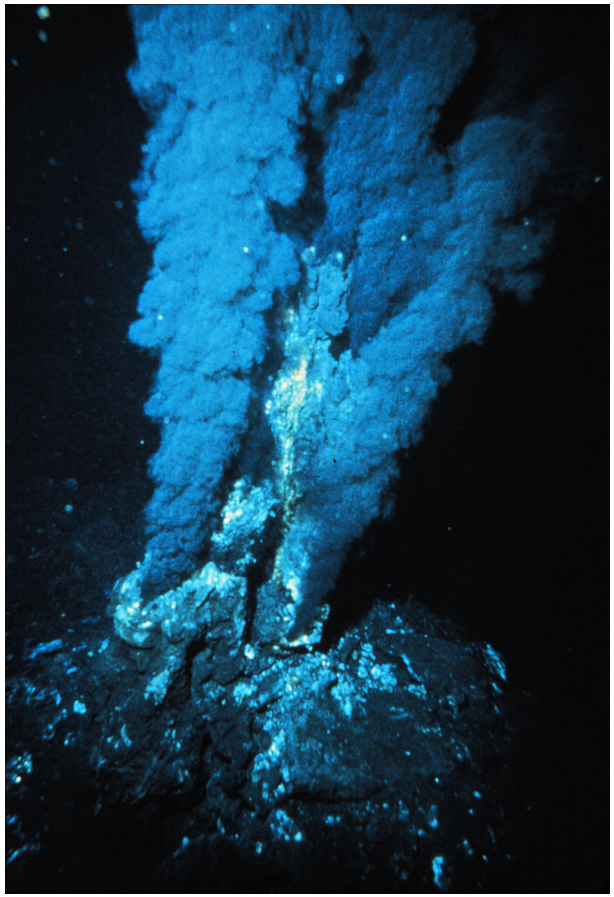
From Ice to Fire: The Incredible Science of Temperature: A Temperature for Life



https://learningonscreen.ac.uk/ondemand/index.php/clip/151474

Full episode:
https://learningonscreen.ac.uk/ondemand/index.php/prog/10B1B3DD?bcast=126189126

The Origins of Life – making *Order* out of *Chaos*

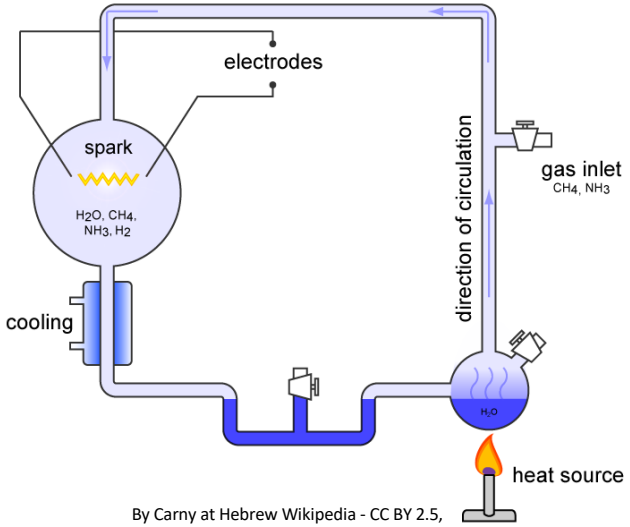


https://commons.wikimedia.org/wiki/File%3ABlacksmoker_in_Atlantic_Ocean.jpg



Stanley Miller

1953 - Urey-Miller Experiment



By Carny at Hebrew Wikipedia - CC BY 2.5, https://commons.wikimedia.org/w/index.php?curid=2173230

https://en.wikipedia.org/wiki/Miller%E2%80%93Urey_experiment

The Urey-Miller Chemical Cocktail…

$CO_2 \rightarrow CO + [O]$

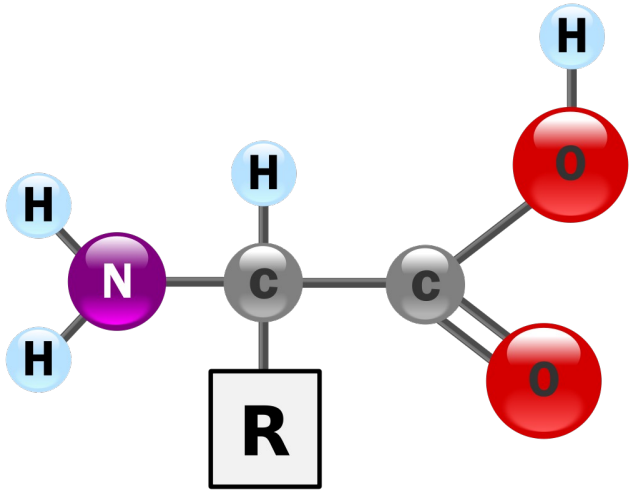
$CH_4 + 2[O] \rightarrow CH_2O + H_2O$ (formaldehyde)

$CO + NH_3 \rightarrow HCN + H_2O$ (cyanide)

$CH_4 + NH_3 \rightarrow HCN + 3H_2$

$CH_2O + HCN + NH_3 \rightarrow NH_2\text{-}CH_2\text{-}CN + H_2O$ (aminoacetonitrile)

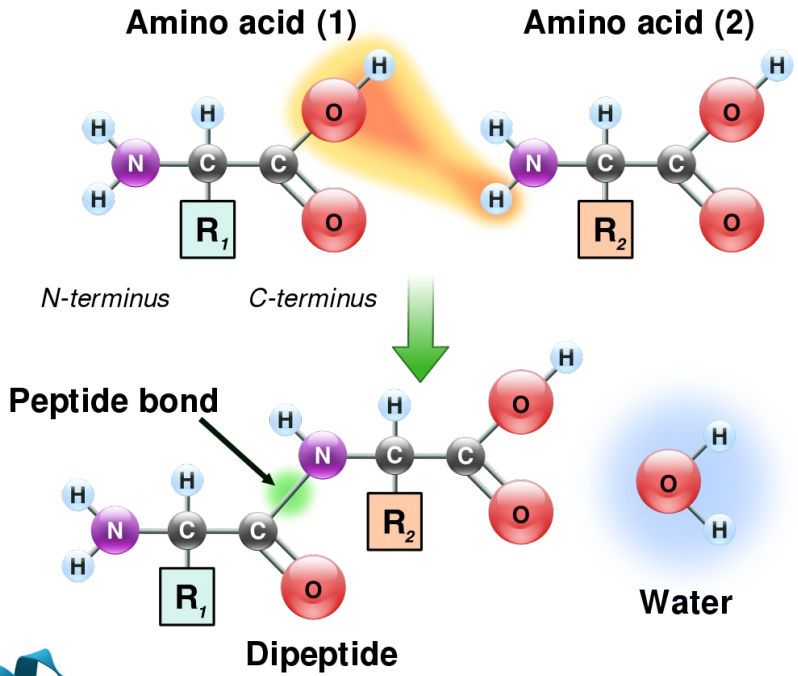
$NH_2\text{-}CH_2\text{-}CN + 2H_2O \rightarrow NH_3 + NH_2\text{-}CH_2\text{-}COOH$ (glycine)



If R = hydrogen:

Glycine – the simplest amino acid

Proteins are chains of amino acids

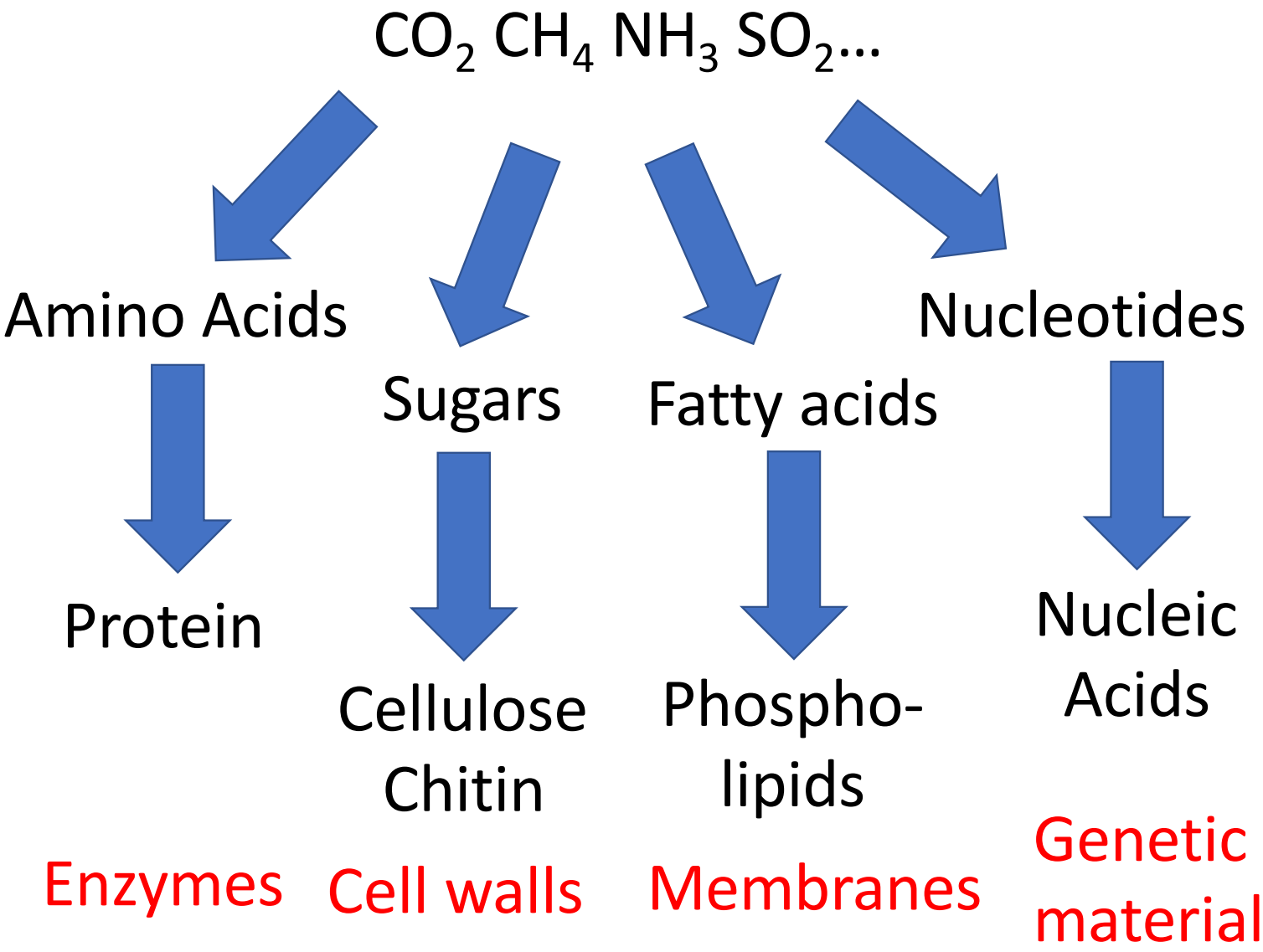




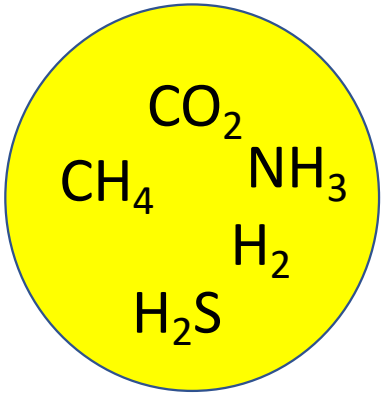
Myoglobin

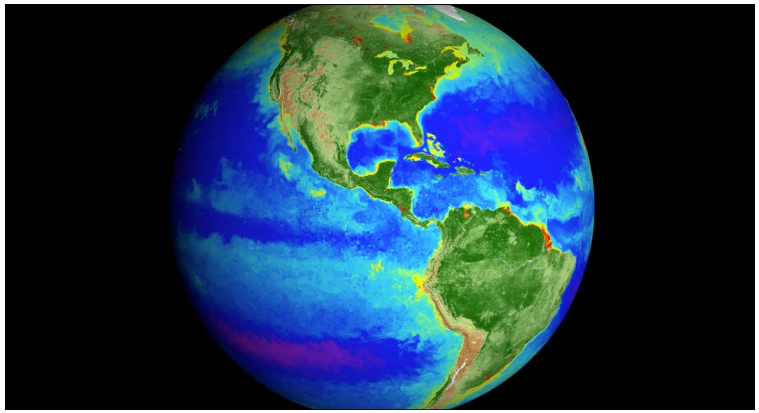
By →AzaToth - self made based on PDB entry, Public Domain, https://commons.wikimedia.org/w/index.php?curid=68596

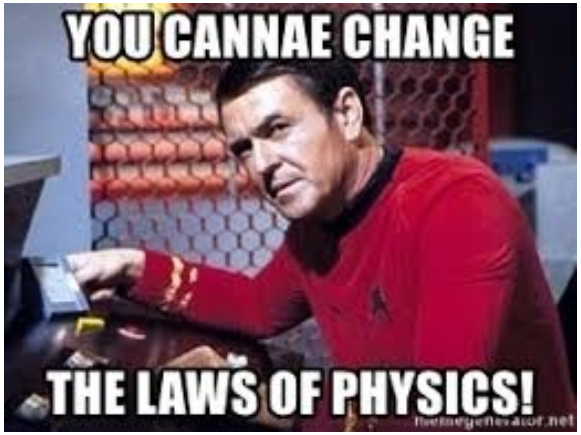
By GYassineMrabetTalk✉This W3C-unspecified vector image was created with Inkscape. - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=2552310


CO_2 CH_4 NH_3 SO_2…
Amino Acids
Sugars
Fatty acids
Nucleotides
Protein
Cellulose
Chitin
Phospho-
lipids
Nucleic
Acids
Enzymes
Cell walls
Membranes
Genetic
material

Simple chemicals can form complex structures needed for life







Needs Energy

Harnessing Energy

ATP – the universal energy currency of the cell

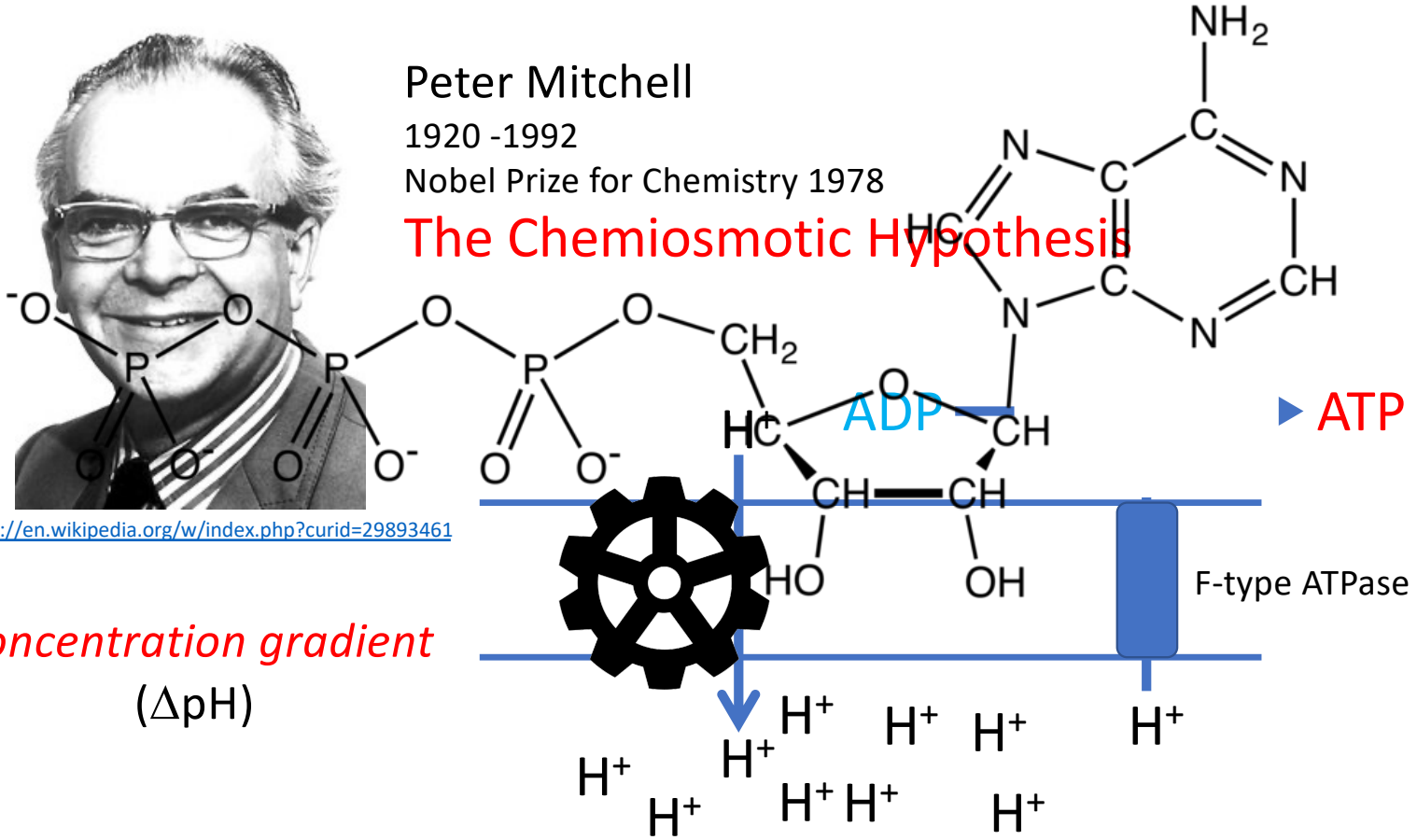
Peter Mitchell

1920 -1992

Nobel Prize for Chemistry 1978

https://en.wikipedia.org/w/index.php?curid=29893461

The Chemiosmotic Hypothesis

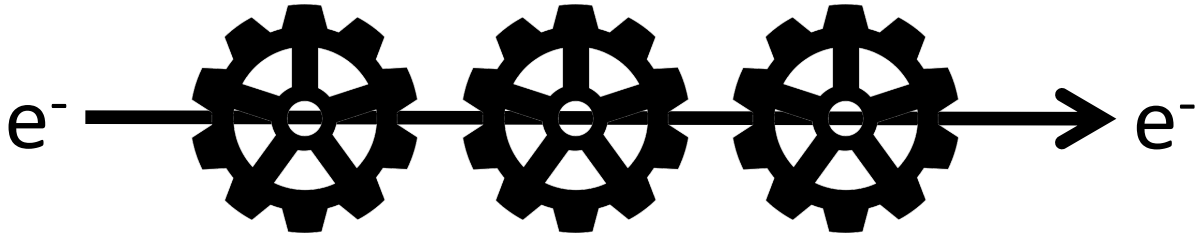


Concentration gradient

(ΔpH)

Ubiquitious – already present in LUCA

Making a pH gradient…



Electron transport chain

Electron transport chains in Archaea and bacteria –
Feature of LUCA

Needs electron *donors* (reductants)
And electron *acceptors* (oxidants)

REDOX

Chemoautotrophs

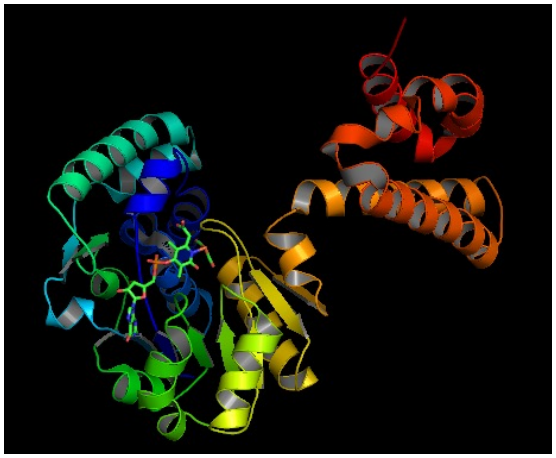
Use chemicals from the environment as reductants –

Chemical Energy

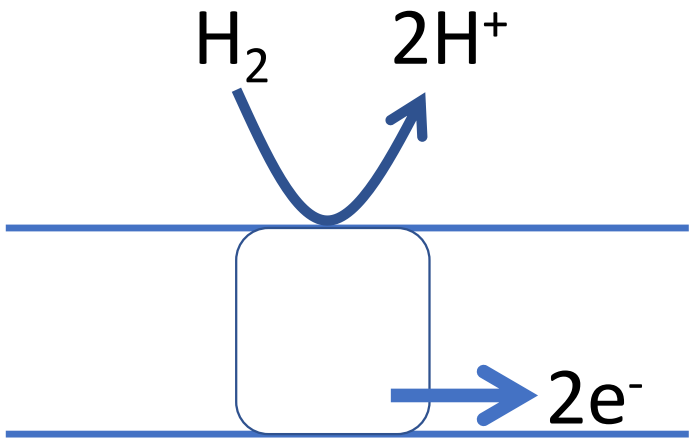
Reductants: e.g. Fe^{2+}; H_2S; S; H_2; NH_3…

All high energy, chemically reactive

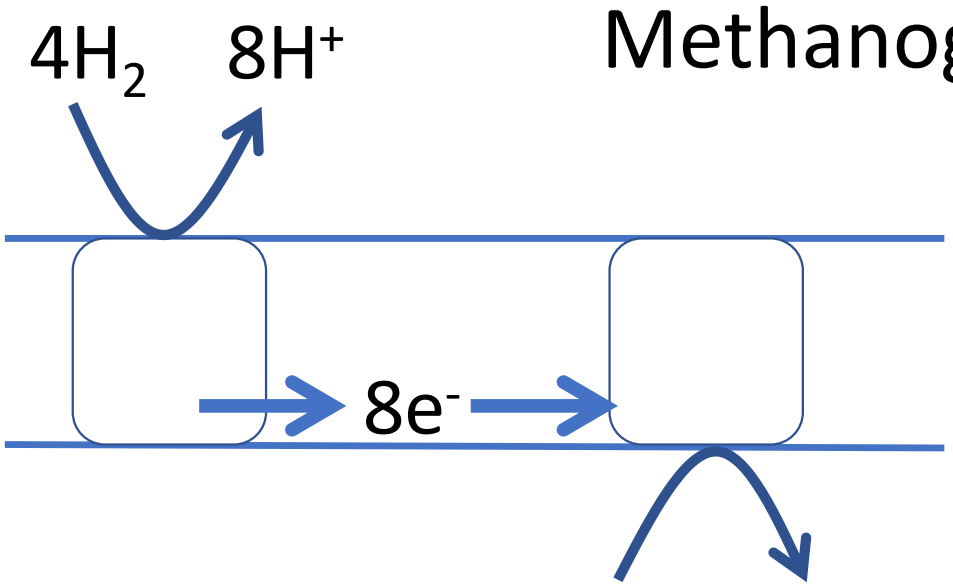
e.g. Hydrogen…



Hydrogenase



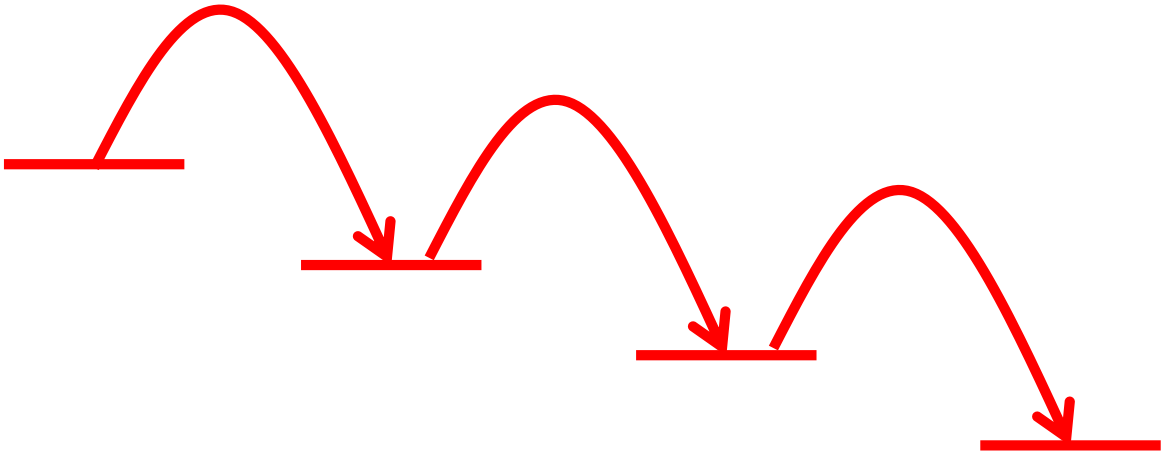
Methanogenesis





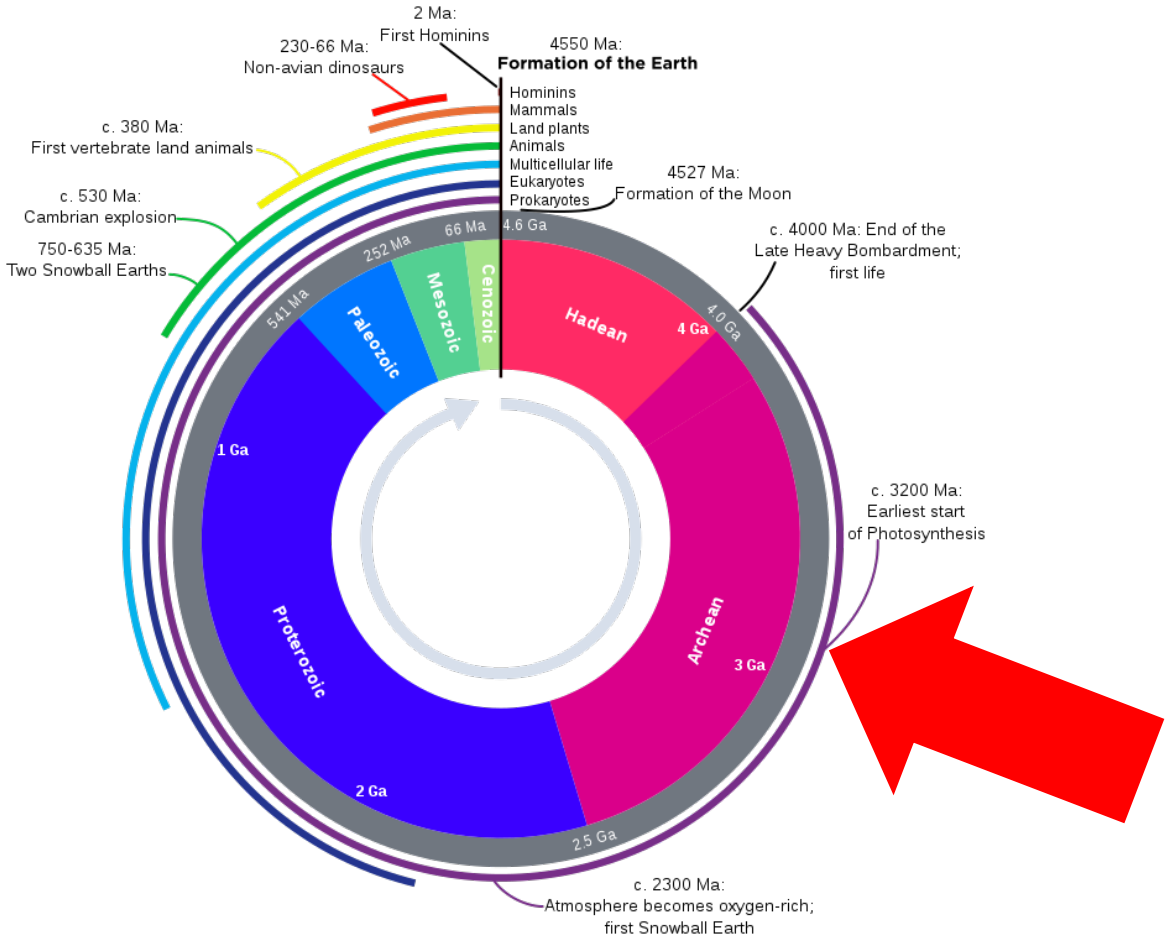
The Will o' the Wisp and the Snake by Hermann Hendrich (1854-1931
https://commons.wikimedia.org/w/index.php?curid=32916743

Chemotrophs rely on a world out of equilibrium

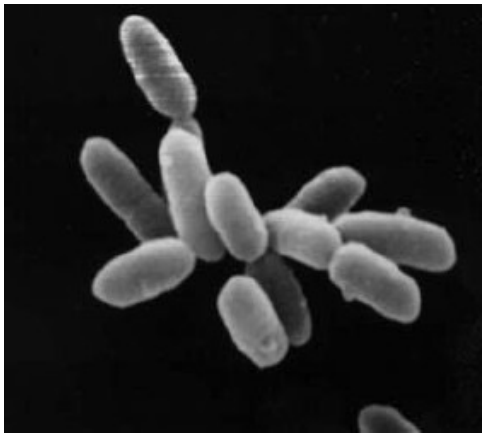


Reactions must overall be exothermic

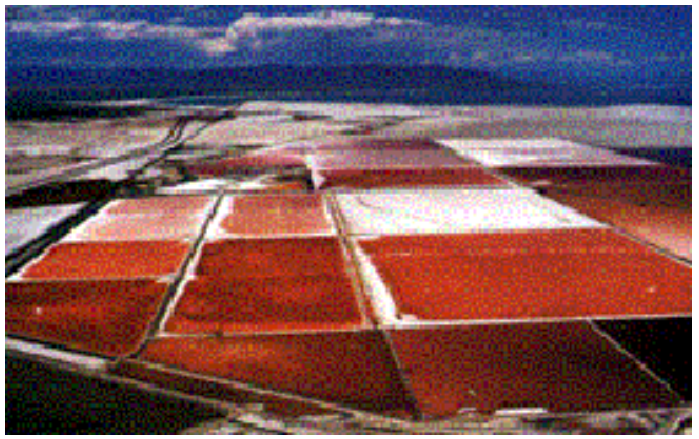
Let there be light



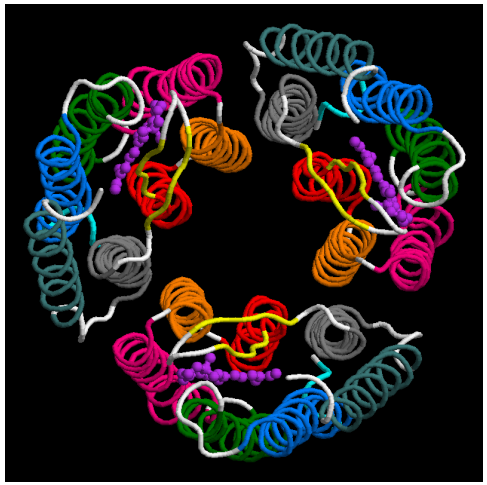
Archaean Photosynthesis



By NASA Public Domain,
https://commons.wikimedia.org/w/index.php?curid=2979987

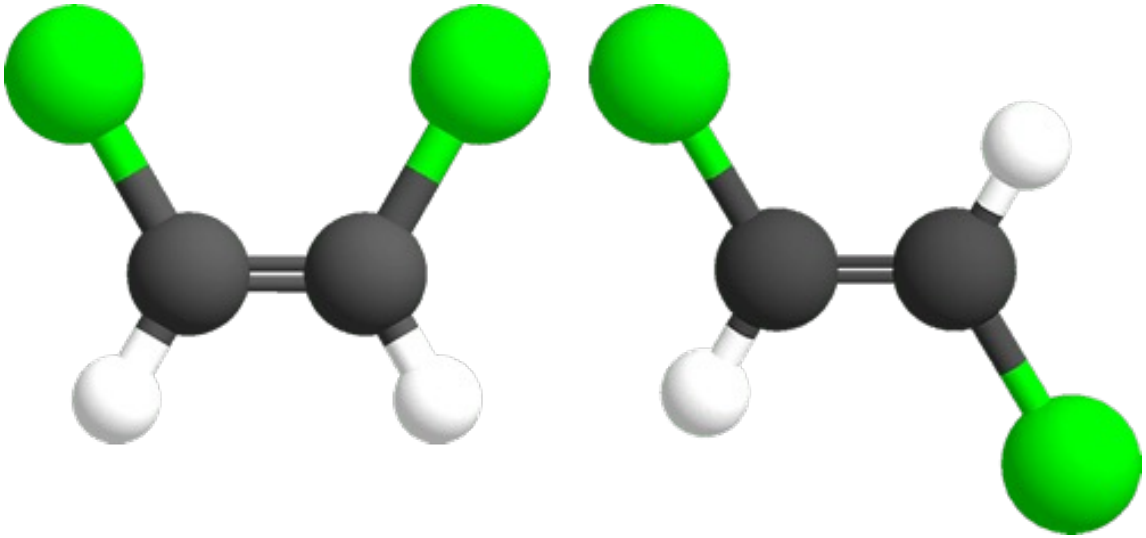


e.g. *Halobacterium salinarum*

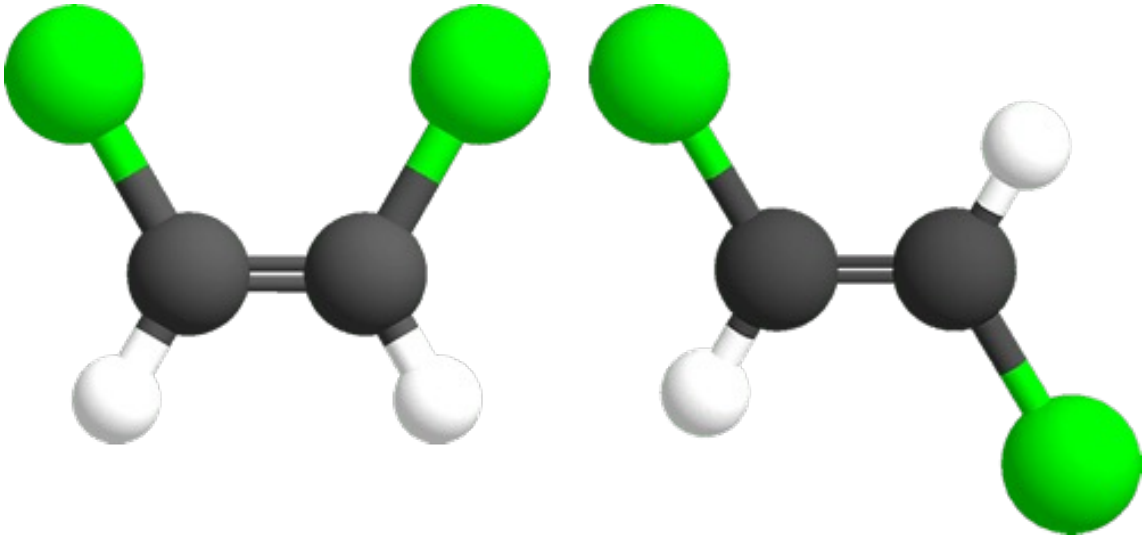


BUT first, a bit of chemistry…

Isomerisation



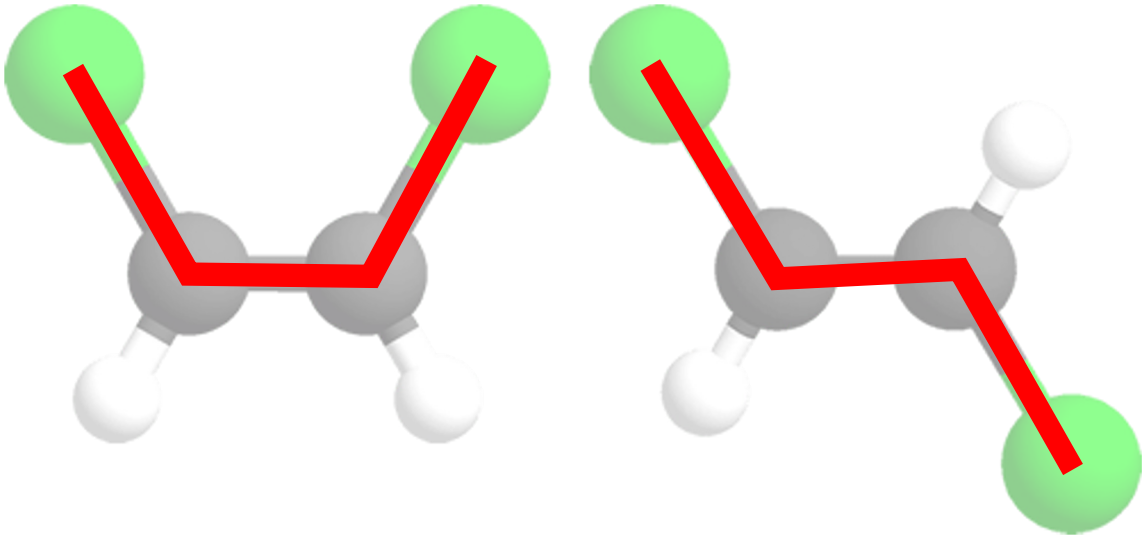
BUT first, a bit of chemistry…



cis-isomer

trans-isomer

BUT first, a bit of chemistry…



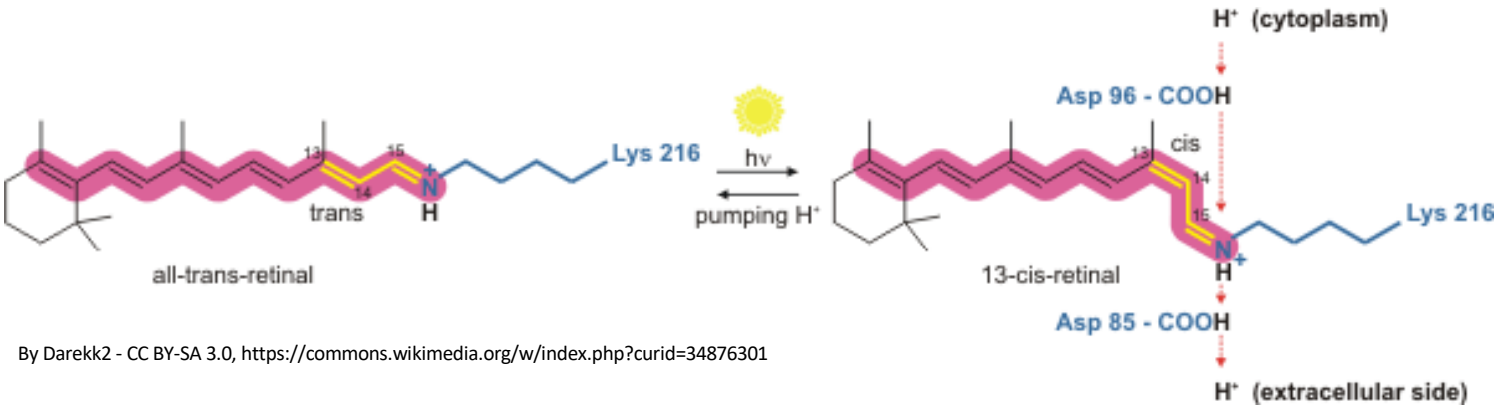
cis-isomer

‘Boat’

trans-isomer

‘Seat’

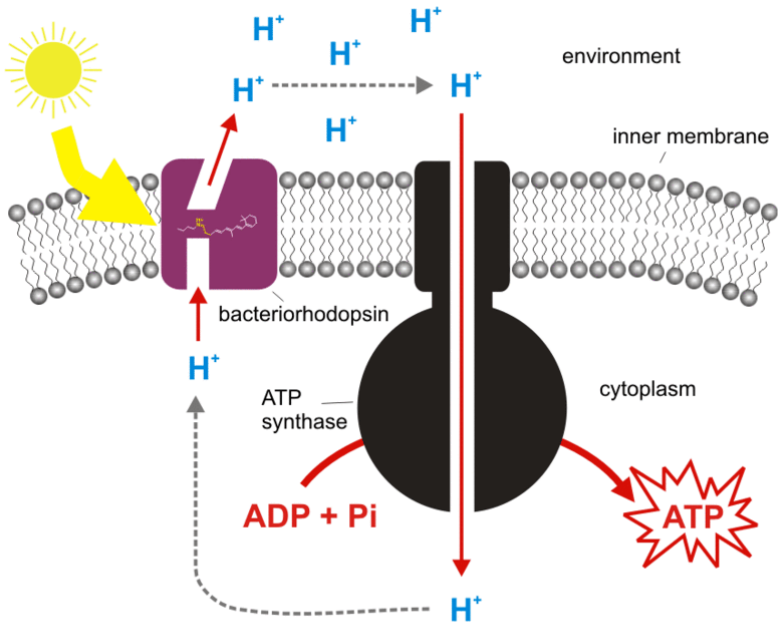
Bacteriorhodopsin - retinal

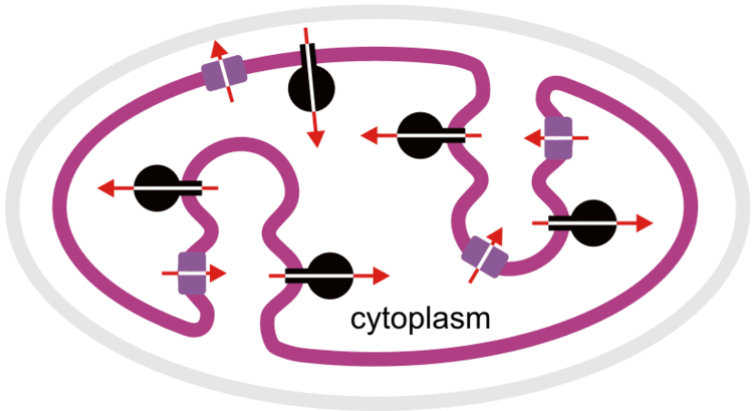


By Darekk2 - CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=34876301

Cis-trans isomerisation driven by light – 2 H^+ per photon of light
Pumps protons but no electrons moved

Absorb BLUE and GREEN light, so the colour is…



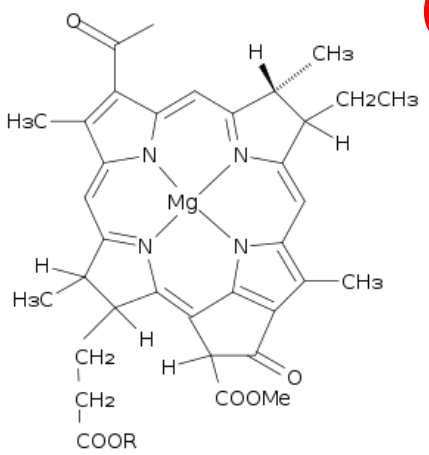


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By Aussie Oc - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=42372752

Bacterial photosynthesis

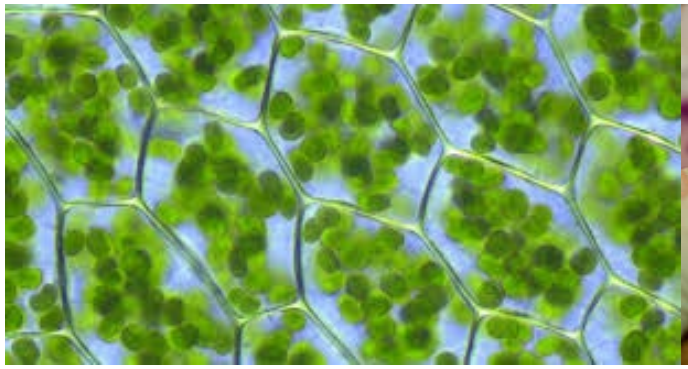
Chlorophyll



Multiple forms (a-g)

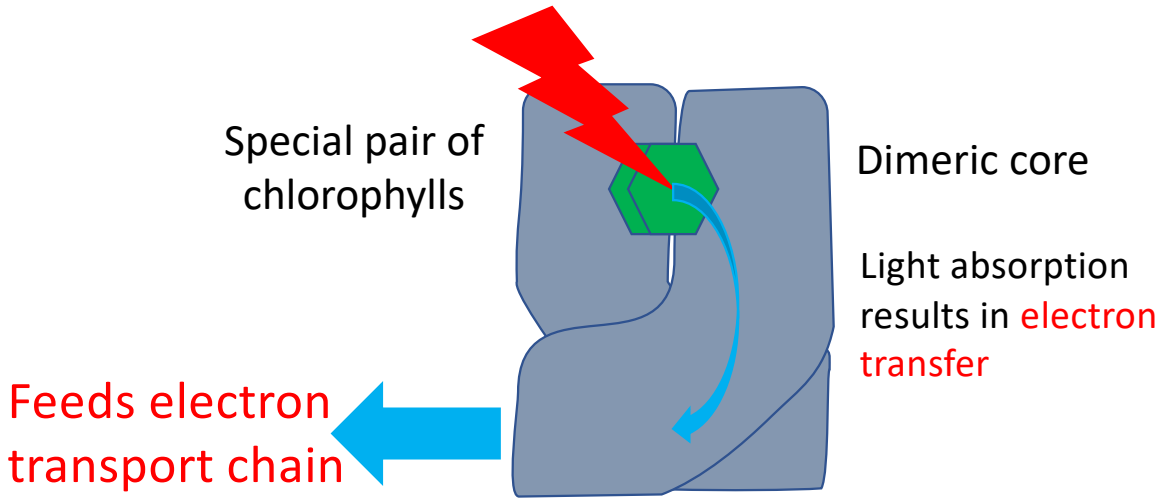
Absorb blue light and either red or near infrared

So, colour is….?





Reaction centres



Turning light energy into chemical (redox) energy

Key points so far…

- Organic molecules formed spontaneously on the early Earth
- Life somehow happened
- Organisms alive today all evolved from the Last Universal Common Ancestor (LUCA)
- Life needs to capture energy to form complex molecules
- Most modern life captures energy from light via photosynthesis

Next week: Oxygen – how one tiny organism changed the planet forever

Before next week: Please review this weeks materials (approx. 2 hours)
Watch and evaluate Astrum podcast (approx. 2 hours)
: https://youtu.be/C3hBF3Gbj8Y?si=M1JPgxBNpfpEg8gj

Next time: Oxygen – how one tiny organism changed the planet forever

Further reading:

Further reading: “Life: the Science of Biology” (Sadava et al.) – available via Blackboard

Chapter 1.1

Chapters 9 & 10

Section 9.3 – Chemiosmosis

Section 10.2 - Photosynthesis

(For further background:
Lane, N. (2010) Why Are Cells Powered by Proton Gradients? Nature Education 3(9):18)